Product Requirement Document (PRD) — CliniFlow AI

Team Name : Dependency Free

Participant Name : Tang Shao Xian

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1. Project Overview

CliniFlow AI is an AI-powered clinical workflow platform designed to streamline and automate the end-to-end patient journey for small and medium-sized (SME) clinics in Malaysia. The system integrates intelligent agents across the **pre-visit**, **visiting**, and **post-visit** phases of every consultation, with a dedicated **safety evaluator** that screens AI-generated clinical content before it reaches the doctor — reducing manual workload, improving documentation quality, and protecting patient safety.

The platform is built to support every stakeholder inside a clinic ecosystem. **Patients** interact with an AI-driven symptom-intake agent before their visit, enabling structured data collection and better preparation for consultations. During the consultation, **clinic doctors** use the system to automatically generate structured medical documentation (SOAP notes) from text or voice input, significantly reducing time spent on administrative work. **Clinic staff and receptionists** benefit from streamlined intake, walk-in registration, and a real-time waiting list, while **clinic administrators or owners** gain operational oversight through user management, role-based access control, and analytics.

By transforming unstructured clinical interactions into structured, actionable outputs, CliniFlow AI minimises inefficiencies caused by fragmented systems and repetitive workflows. The platform delivers a seamless intelligent workflow that connects every stakeholder, ensuring information flows consistently from initial patient intake through to post-visit follow-up.

Ultimately, CliniFlow AI aims to empower SME clinics with a scalable, AI-driven solution that enhances operational efficiency, improves documentation accuracy, safeguards clinical decisions, and elevates the overall patient experience.

Key Stakeholders:

Role	Description
Clinic Doctors	Primary users who conduct consultations and review AI-generated documentation
Clinic Staff / Receptionists	Manage patient intake, walk-in registration, check-in, and the daily waiting list
Patients	Interact with the symptom-intake agent, book appointments, and review post-visit records

Clinic Administrators / Owners	Oversee operations, manage user accounts, configure roles, and view analytics

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2. Background & Business Objective

2.1 Background

SME clinics in Malaysia face significant operational challenges due to high patient volumes, limited resources, and reliance on manual processes.

- **Heavy administrative burden:** Doctors spend a large portion of their time on documentation rather than patient care. Studies show clinicians can spend **30–50% of their time on administrative tasks**, which reduces consultation quality and contributes to burnout.
- **Fragmented systems:** Many clinics still rely on a mix of paper records and basic digital tools, leading to inconsistent, duplicated, or incomplete patient data. Unlike large hospitals, SME clinics often lack integrated electronic medical record systems.
- **Repetitive data collection:** Patients are frequently asked to repeat the same information (symptoms, history, allergies) at different stages, causing inefficiency and frustration.
- **Manual safety checks:** Doctors check drug–drug interactions, allergy conflicts, and dosing from memory or scattered references, leaving room for missed contraindications — especially under time pressure.
- **Poor patient communication:** Due to time constraints and varying health literacy levels in Malaysia, patients often leave consultations without fully understanding medication instructions or follow-up care, leading to low adherence and poorer outcomes.

Overall, these challenges highlight the need for a solution that reduces administrative workload, improves data consistency, surfaces clinical-safety risks, and enhances communication between doctors and patients.

2.2 Business Objectives

	Objective	Success Metric
1	Reduce documentation time for doctors	≥ 50% reduction in time spent on clinical note-taking per consultation
2	Improve data accuracy and completeness	≥ 90% of patient records contain all required structured fields after each visit
3	Enhance patient understanding	≥ 80% of patients report clear understanding of post-visit instructions (via survey)
4	Streamline clinic workflows	≥ 40% reduction in average patient check-in to consultation-ready time
5	Centralise patient records	100% of patient records accessible from a single platform within 6 months of deployment
6	Reduce clinical-safety risk	≥ 95% of generated prescriptions screened for drug–drug interactions, allergies, and dosing before doctor sign-off

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3. Product Purpose

Current manual workflow in clinics:

> Patient arrives with no preparation → fills paper form with generic questions → form is illegible or incomplete → doctor re-asks everything → doctor examines while typing notes (30–50% of consult on paperwork) → manually checks drug interactions from memory → writes paper prescription → patient leaves with rushed verbal explanation → forgets or misunderstands instructions → low adherence, repeat visits.

CliniFlow AI serves as an **end-to-end intelligent workflow agent** that bridges the gap between clinical care and administrative efficiency. Its purpose is to:

1. **Collect and structure patient data before consultation** — CliniFlow AI enables structured data collection during the pre-visit phase through an AI-driven symptom-intake agent. Instead of relying on unstructured or incomplete patient descriptions, the system guides patients to provide relevant information and organises it into a clear, standardised format the doctor can review before the visit.
2. **Automate clinical documentation during consultation** — The platform assists doctors by automatically generating structured medical documentation (SOAP notes) from consultation inputs, including text and voice recordings.
3. **Screen AI output for clinical-safety risk** — Before the doctor reviews the generated note, an evaluator agent checks the prescription against the patient's allergies, current medications, weight, age, and pregnancy status, flagging any drug–drug interactions, allergy conflicts, dosing concerns, or missing information.
4. **Ensure consistent and structured medical records** — CliniFlow AI centralises patient information into a unified system, ensuring that all records are stored in a structured and consistent format and accessible to authorised roles.
5. **Improve patient understanding and post-visit care** — After the consultation, the system generates clear, simplified instructions for patients, including medication details, dosage, and care guidelines, delivered through both the patient portal and a messaging channel.

Value Proposition:

- **For Doctors:** Less time documenting, more time caring. AI handles the paperwork, and the safety evaluator catches things the doctor's memory might miss.
- **For Patients:** Better understanding of their health, clearer instructions, fewer repeated questions, and a more seamless clinic experience.
- **For Clinic Staff:** A single source of truth for who is waiting, who is checked in, and what's next — no more clipboards.
- **For Clinic Owners:** Improved operational efficiency, reduced administrative costs, audit-ready records, and higher patient satisfaction.

4. System Functionalities

The system is divided into three patient-journey phases plus core platform features.

4.1 Pre-Visit Phase: AI Symptom Intake Agent

Feature	Description
Conversational Symptom Collection	AI chatbot engages patients in a guided conversation to collect symptoms, medical history, allergies, and current medications before the consultation. Text and voice input are both supported.
Multi-language Support	Supports English and Bahasa Malaysia for patient interaction.
Structured Pre-Visit Report	Automatically generates a structured symptom summary report for the doctor to review before the consultation.
Patient Profile Auto-Population	Pre-fills known patient information from previous visits to avoid repetitive data entry.
Appointment Booking	Patients book an appointment slot from a 14-day rolling availability calendar after completing intake.
PDPA Consent Capture	Explicit consent for data processing is captured and recorded before any clinical data is collected.

4.2 Visiting Phase: AI Clinical Documentation

Feature	Description
Multi-Modal Input	Supports both text-based note entry and voice recording during consultations, allowing flexibility based on doctor preference and workflow.
Speech-to-Text Transcription	Converts doctor-patient conversations into text, enabling automatic capture of consultation details for further processing.
Real-Time / Post-Consultation Processing	Allows documentation to be generated either in real-time during consultation or after the session using recorded inputs.
AI-Structured Clinical Notes	Automatically organises transcribed or typed inputs into the SOAP (Subjective, Objective, Assessment, Plan) format to standardise documentation.
Medical Entity Extraction	Identifies and extracts key clinical information such as symptoms, duration, medications, and observations from unstructured inputs.
Missing Information Detection	Detects incomplete or missing critical information (e.g., allergies, duration of symptoms) and prompts the user to provide additional details.
Clinical-Safety Evaluator	Screens AI-generated prescriptions for drug–drug interactions, allergy conflicts, dosing concerns, pregnancy contraindications, hallucinated drugs, and missing information. Critical findings must be acknowledged by the doctor with a written reason before the visit can proceed.
Doctor Review & Editing	Allows doctors to review, edit, and validate AI-generated documentation before finalising records. UI visibly distinguishes AI draft from human-confirmed content.
Centralised Patient Records	Stores all consultation data in a unified patient record system accessible to authorised clinic staff and patients.
Data Structuring & Standardisation	Ensures all captured data follows a consistent structure, improving readability, retrieval, and future reference.

4.3 Post-Visit Phase: Patient Communication & Follow-Up

Feature	Description
Patient-Friendly Summary	AI generates a clear, non-technical summary of the consultation including diagnosis, prescribed medications, dosage instructions, and follow-up schedule.
Medication & Dosage Instructions	Structured medication list with dosage, frequency, duration, and special instructions.
Patient Portal Access	Patients can log in to review their visit history, summaries, and instructions at any time.
Messaging Delivery	The post-visit summary and medication instructions are also delivered to the patient via WhatsApp so they can re-read the instructions at home without logging in.
Follow-Up Reminders	In-app notifications remind the patient of follow-up appointments.

4.4 Core Platform Features

Feature	Description
User Authentication & Role Management	Secure login with role-based access control (Patient, Doctor, Clinic Staff, Clinic Admin).
Patient Records Management	CRUD operations for patient demographics, medical history, and visit records, with server-side ownership checks on every action.
Staff Workspace	Today's waiting list, patient check-in, walk-in registration, and visit-status visibility for receptionists.
Admin Workspace	User management (create, edit role, deactivate, force password reset), clinic-wide analytics (KPIs, sparklines), and audit-log review for the clinic owner.
Dashboard & Analytics	Overview of clinic operations including patient volume, consultation times, and documentation metrics.

PDPA Audit Log	Append-only audit log records every create/update/delete of per-patient data for compliance review.

5. User Stories & Use Cases

5.1 Patient User Stories

ID	User Story	Acceptance Criteria
US-P01	As a patient, I want to describe my symptoms before visiting so that the doctor understands my condition in advance	• Patient can input symptoms in text form • System accepts unstructured input • Input is successfully submitted and processed
US-P02	As a patient, I want the system to ask follow-up questions so that my symptoms are captured accurately	• System detects missing information (e.g., duration, severity) • Follow-up questions are generated dynamically • Patient can respond and update information
US-P03	As a patient, I want to book an appointment after completing intake so that I know when to come in	• Patient sees a 14-day rolling availability calendar • Patient can pick a slot and confirm • Booking is reflected in the doctor's and staff's view
US-P04	As a patient, I want to view my medical record after the visit so that I understand what was discussed	• Patient can access their record • Record displays structured consultation details • Data is readable and complete

US-P05	As a patient, I want to receive clear medication instructions so that I can follow treatment correctly	• System generates medication details • Dosage and frequency are included • Instructions are presented in simple language
US-P06	As a patient, I want to receive my summary on a channel I already use so that I don't lose it	• Post-visit summary is delivered via the patient's chosen messaging channel • Same content is available in the portal
US-P07	As a patient, I want to understand my care instructions so that I can follow up properly	• Post-visit summary is generated • Instructions are clear and concise • Key actions (e.g., follow-up) are included

5.2 Doctor User Stories

D	User Story	Acceptance Criteria
US-D01	As a doctor, I want to receive a structured pre-visit report so that I can prepare before consultation	• System generates a structured report (symptoms, duration, history) • Report is accessible before consultation • Report format is clear and readable
US-D02	As a doctor, I want to record consultation details using text or voice so that I can capture information efficiently	• System supports text input • System accepts voice recording • Input is successfully stored
US-D03	As a doctor, I want the system to convert conversations into structured	• Speech/text input is processed by the system

	notes so that I don't need to write manually	• Structured notes (SOAP format) are generated • Output is displayed clearly
US-D04	As a doctor, I want to review and edit AI-generated notes so that I can ensure accuracy before saving	• Generated notes are editable • Doctor can modify content • Final version can be saved • UI visibly distinguishes AI draft from confirmed content
US-D05	As a doctor, I want the system to detect missing information so that important details are not overlooked	• System identifies missing fields (e.g., allergies) • System prompts user for additional input • Updated data is reflected in final report
US-D06	As a doctor, I want to be warned about drug interactions, allergies, and dosing concerns before I sign off so that I don't miss a contraindication	• Evaluator findings appear before finalisation • Critical findings block sign-off until acknowledged • Acknowledgement requires a written reason
US-D07	As a doctor, I want patients to have access to clear instructions so that treatment adherence improves	• Instructions are automatically generated after consultation • Patients can access them via the platform

5.3 Clinic Staff / Receptionist User Stories

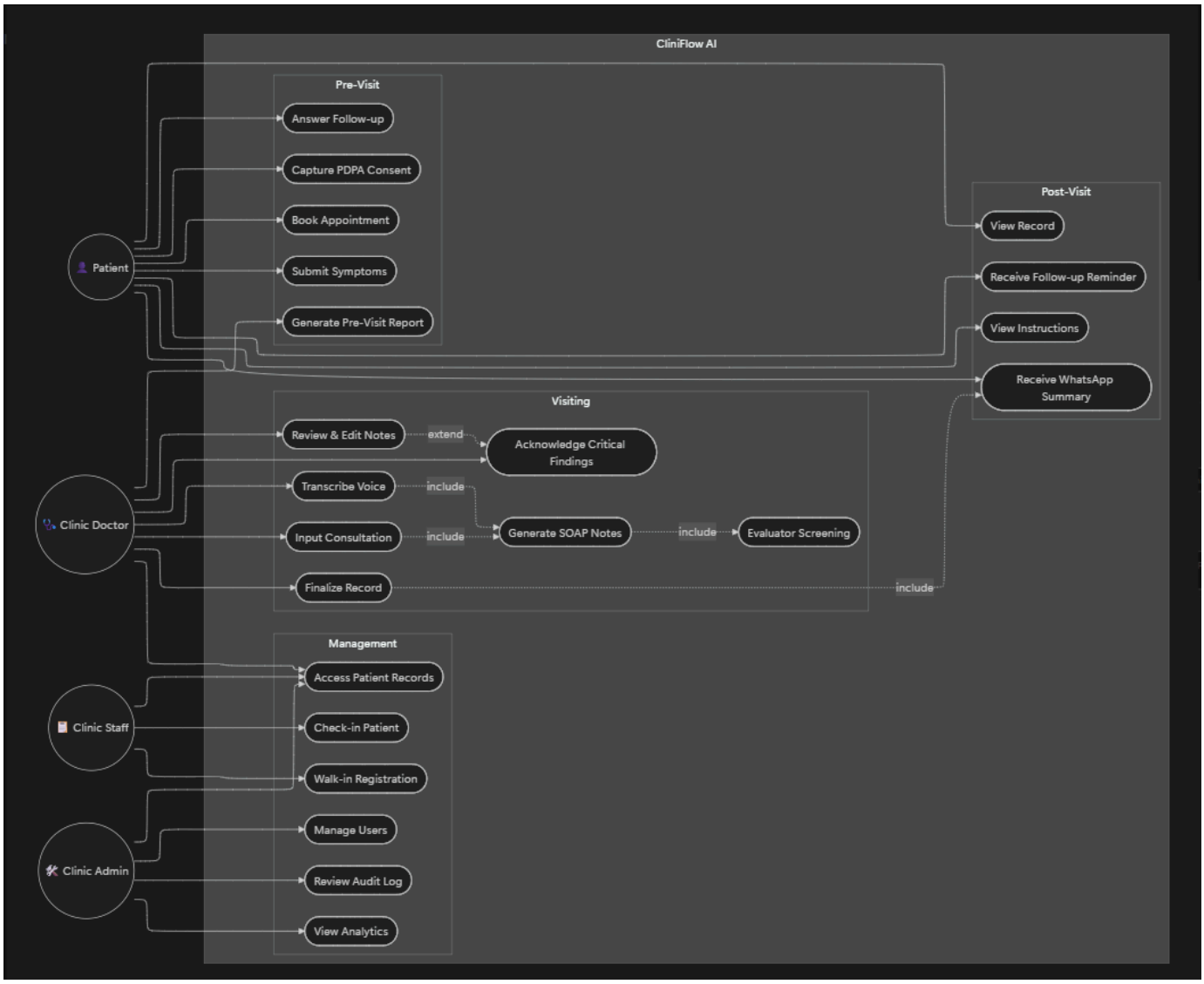
ID	User Story	Acceptance Criteria
US-R01	As clinic staff, I want patient information to be collected before the visit so that intake is faster and more efficient	• Pre-visit data is available before appointment • Data is structured and complete • Staff can access patient information
US-R02	As clinic staff, I want to register a walk-in patient so that they can be seen the same day	• Staff can create a new patient record from the staff workspace • Walk-in is added to today's waiting list
US-R03	As clinic staff, I want to check a patient in when they arrive so that the doctor knows who is ready	• Check-in updates the visit status • Doctor's view reflects the change in real time
US-R04	As clinic staff, I want access to patient records so that I can assist in administrative tasks	• Staff can retrieve patient records • Data is displayed in structured format • Access is limited to authorised users

5.4 Clinic Administrators / Owners

ID	User Story	Acceptance Criteria
US-O01	As an administrator, I want a centralised system for patient records so that clinic operations are more organised	• All patient data is stored in one system • Records are consistent and structured • Data is retrievable when needed
US-O02	As an administrator, I want controlled access to the system so that patient data is secure	• Only authorised users can access records • Role-based access control is implemented • Unauthorised access is restricted
US-O03	As an administrator, I want to manage user accounts so that I can onboard, deactivate, or reset staff as needed	• Admin can create users • Admin can change roles, deactivate accounts, force a password reset • Changes take effect immediately
US-O04	As an administrator, I want a view of clinic activity so that I can understand operational health	• Dashboard shows daily patient volume, consultation times, and key KPIs • Trends are visible over time
US-O05	As an administrator, I want an audit trail so that I can demonstrate PDPA compliance	• Every create/update/delete on per-patient data is recorded • Audit log is append-only

5.5 Use Case Diagram

The system covers four primary actor groups (Patient, Clinic Doctor, Clinic Staff, Clinic Admin) interacting with four functional areas (Pre-Visit, Visiting, Post-Visit, Management).



Relationships:

- *Generate SOAP Notes* is included whenever consultation input is captured (text or voice).
- *Evaluator Screening* is always included in *Generate SOAP Notes* before the doctor reviews.

- *Acknowledge Critical Findings* extends *Review & Edit Notes* — only triggered when the evaluator returns a critical finding.
- *Finalize Record* includes the *Receive WhatsApp Summary* trigger so the patient gets the post-visit message automatically.

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6. Features Included (Scope Definition)

6.1 Pre-Visit

	Features	Priority
1	Symptom Intake Agent	Must have
2	Pre-visit report generation	Must have
3	PDPA consent capture	Must have
4	Appointment booking (14-day rolling availability)	Must have
5	Voice input in symptom collection	Should have
6	Bahasa Malaysia interaction	Should have

6.2 Visiting

	Features	Priority
1	Multi-input support (text + voice)	Must have
2	Documentation Agent (SOAP)	Must have
3	Structured report generation	Must have
4	Patient record storage	Must have
5	Doctor review & confirm gate (AI never auto-finalises)	Must have
6	Clinical-safety evaluator (DDI, allergy, dose, pregnancy, hallucination, completeness)	Must have
7	AI-suggested diagnosis codes and medication auto-complete	Should have

6.3 Post-Visit

	Features	Priority
1	Patient record viewing	Must have
2	Instruction generation (medication, dosage, etc.)	Must have
3	Patient-friendly summary	Must have
4	Messaging delivery of summary (WhatsApp)	Should have
5	Follow-up appointment reminders (in-app notifications)	Should have
6	Medication schedule reminders	Should have

6.4 Core Platform

	Features	Priority
1	Authentication & role-based access control	Must have
2	Patient record CRUD with ownership checks	Must have
3	Append-only PDPA audit log	Must have
4	Staff workspace (waiting list, walk-in, check-in)	Must have
5	Admin workspace (user management)	Must have
6	Operational analytics dashboard for clinic admins	Should have

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7. Features Not Included (Scope Control)

The following features are explicitly **out of scope** for this project to maintain focus and feasibility:

	Excluded Feature	Reason
1	E-prescribing with direct pharmacy integration	Requires regulatory compliance with Malaysian Pharmacy Board; deferred to future phase.
2	Telemedicine / video consultation	Adds significant complexity in infrastructure and compliance; not part of core workflow automation.
3	Insurance claims processing	Requires integration with Malaysian insurance providers; separate initiative.
4	Medical imaging analysis (X-ray, MRI)	Specialised AI domain outside the scope of workflow automation.
5	Full Electronic Health Record (EHR) system	CliniFlow AI focuses on workflow automation, not replacing full EHR systems; integration with existing EHRs may be considered later.
6	Billing and payment processing	Separate domain; clinics typically have existing billing systems.
7	Mobile native app (iOS / Android)	Web-responsive design is prioritised; native apps may be considered in future phases.
8	Multi-clinic / multi-tenant management	MVP targets a single-clinic deployment per instance; multi-tenant orchestration is a future phase.

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8. Assumptions & Constraints

8.1 Assumptions

	Assumption
1	Target clinics have reliable internet connectivity for web-based access.
2	Doctors and staff have basic digital literacy and can interact with a web application.
3	Patients have access to a smartphone or computer to interact with the symptom-intake chatbot and patient portal.
4	Clinics are willing to adopt a new digital workflow tool alongside or replacing their current manual processes.
5	AI-generated clinical notes will always require doctor review and confirmation before being finalised — the AI assists but does not replace clinical judgment.
6	English and Bahasa Malaysia are sufficient for MVP; additional languages (Mandarin, Tamil) are deferred to a later phase.
7	Voice transcription accuracy of ≥ 90% is achievable for Malaysian-accented English using available speech-to-text APIs.
8	Patients have access to a messaging channel (WhatsApp) to receive their post-visit summary.
9	The clinic owner is the data controller under PDPA and is responsible for collecting patient consent.

8.2 Constraints

	Constraint	Impact
1	**Data Privacy Compliance:** Must comply with Malaysia's Personal Data Protection Act 2010 (PDPA) for handling patient health information.	Affects data storage, access controls, consent mechanisms, and requires an append-only audit trail.
2	**AI Accuracy Limitations:** AI-generated notes and summaries must be treated as drafts requiring human verification; the system must not imply autonomous clinical decision-making.	Requires clear UI indicators that AI output is a suggestion, not a diagnosis, and a mandatory doctor confirm step.
3	**Clinical-Safety Gate:** Critical evaluator findings (drug–drug interactions, allergies, dosing, pregnancy) must be acknowledged by the doctor before a visit can be finalised.	Adds a UX gate before sign-off; cannot be auto-bypassed; requires a written acknowledgement reason.
4	**Budget Constraints:** SME clinics have limited IT budgets; solutions must be cost-effective with minimal infrastructure requirements.	Cloud-hosted SaaS model preferred; avoid expensive on-premise deployments.
5	**Timeline:** MVP must be deliverable within 3 months to validate the core value proposition with pilot clinics.	Limits Phase 1 scope to essential features only.
6	**Language / Accent Diversity:** Malaysia has diverse linguistic patterns; voice transcription must handle code-switching (English / Malay mix).	May limit voice feature accuracy; requires iterative improvement.
7	**AI / LLM Dependency:** The platform depends on third-party LLM and speech-to-text providers.	Requires abstraction behind an internal API layer, fallback handling, and cost monitoring.

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9. Risks & Questions Throughout Development

9.1 Risks

	Risk	Likelihood	Impact	Mitigation Strategy
R1	AI-generated clinical notes contain inaccuracies that go undetected by the reviewing doctor.	Medium	High	Mandatory doctor review step with explicit confirmation; highlight AI-generated sections; clinical-safety evaluator screens prescriptions before review; audit trail for all changes.
R2	Low adoption by doctors who are accustomed to manual workflows or resistant to AI tools.	Medium	High	Conduct user research and co-design sessions with pilot clinic doctors; provide comprehensive onboarding and training; iterate based on feedback.
R3	Voice transcription accuracy is insufficient for Malaysian-accented English and code-switching.	Medium	Medium	Start with text-based input as a fallback; evaluate multiple speech-to-text providers; collect voice samples for fine-tuning.
R4	Patient data breach or non-compliance with PDPA.	Low	Critical	Implement encryption at rest and in transit; role-based access control; append-only audit log; regular security audits; legal review of data-handling practices.
R5	Scope creep due to clinic-specific customisation requests during pilot.	High	Medium	Clearly define MVP scope (Section 6); use a feature-request backlog; evaluate requests against the product roadmap before committing.
R6	Dependency on third-party AI / ML APIs (LLM, speech-to-text)	Medium	Medium	Abstract AI services behind an internal API layer; implement fallback mechanisms; monitor API

	introduces latency, cost, or availability risks.			costs and performance; circuit-breaker on outbound calls.
R7	Evaluator agent produces false positives that block doctor workflow and erode trust.	Medium	Medium	Allow doctors to override critical findings with a written reason; record the override for audit; tune thresholds based on pilot feedback.
R8	Patient consent under PDPA is not captured or recorded properly, exposing the clinic to compliance risk.	Low	High	Capture explicit consent before any data collection; persist consent in the audit log; reject data submission without valid consent.

9.2 Open Questions

	Question	Owner	Status
Q1	What specific PDPA compliance requirements apply to AI-processed health data in Malaysia?	Legal / Compliance	Open
Q2	Which LLM provider offers the best balance of accuracy, cost, and latency for clinical note generation?	Engineering	Open
Q3	What is the preferred SOAP note format used by Malaysian SME clinics? Are there local variations?	Product / Medical Advisor	Open
Q4	How should the system handle cases where AI confidence in generated notes is low?	Product / UX	Open
Q5	What is the acceptable latency for real-time voice transcription during consultations?	Engineering / UX	Open
Q6	Will pilot clinics provide access to anonymised historical patient data for AI model training / testing?	Business Development	Open
Q7	What integration points (if any) exist with clinics' current systems (e.g., appointment scheduling, billing)?	Product / Engineering	Open
Q8	What is the authoritative source for the Malaysia drug knowledge base used by the evaluator (interactions, dosing, contraindications)?	Product / Medical Advisor	Open

Q9	How will the doctor's evaluator-override reasons be reviewed for ongoing clinical-safety auditing?	Product / Compliance	Open